This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Audio Processor
Manufacturer: Audac
Firmware Version: V4.10(Software)
Model(s): R2

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.2

Version History

Module Version	Date	Notes
1_0_0_0	12/22/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.connectionCounter = 5
- Device Address variable must be set accordingly. Default value is '1'. Device Address ranges from '1' to '99'. Example: InterfaceName.DeviceAddress = '1'
- All zone settings will be lost if the device is switched off. To keep the changes you must save them with the “Save” command.
- Recommended time between commands is 0.1 seconds.
- Recommended repeat rate is 0.2 seconds.

Control Commands, States, and Qualifiers:

Save	None		
ZoneBass	-9 to 9 in steps of 1 dB		
['Zone']	'1' – '8'		
ZoneInputRouting	'Direct Input 1'	'Direct Input 2'	'Direct Input 3'
	'Direct Input 4'	'Direct Input 5'	'Direct Input 6'
	'Direct Input 7'	'Direct Input 8'	'Wall Plate Input 1'
	'Wall Plate Input 2'	'Wall Plate Input 3'	'Wall Plate Input 4'
	'Wall Plate Input 5'	'Wall Plate Input 6'	'Wall Plate Input 7'
	'Wall Plate Input 8'	'Fiber Input 1'	'Fiber Input 2'
	'Fiber Input 3'	'Fiber Input 4'	'Fiber Input 5'
	'Fiber Input 6'	'Fiber Input 7'	'Fiber Input 8'
	'Prio 1'	'Prio 2'	'Internal sine generator'
	'Internal white noise generator'	'Internal pink noise generator'	'Digital/Optical'
['Zone']	'1' – '8'	'All'	
ZoneMute	'On'	'Off'	
['Zone']	'1' – '8'	'All'	
ZoneTreble	-9 to 9 in steps of 1 dB		
['Zone']	'1' – '8'		
ZoneVolume	-70 to 0 in steps of 1 dB		
['Zone']	'1' – '8'	'All'	

Status Available:

ConnectionStatus	'Connected'	'Disconnected'	
ZoneBass	-9 to 9 in steps of 1 dB		
['Zone']	'1' – '8'		
ZoneInputRouting	'No Input'	'Other'	'Direct Input 1'
	'Direct Input 2'	'Direct Input 3'	'Direct Input 4'
	'Direct Input 5'	'Direct Input 6'	'Direct Input 7'
	'Direct Input 8'	'Wall Plate Input 1'	'Wall Plate Input 2'
	'Wall Plate Input 3'	'Wall Plate Input 4'	'Wall Plate Input 5'
	'Wall Plate Input 6'	'Wall Plate Input 7'	'Wall Plate Input 8'
	'Fiber Input 1'	'Fiber Input 2'	'Fiber Input 3'
	'Fiber Input 4'	'Fiber Input 5'	'Fiber Input 6'
	'Fiber Input 7'	'Fiber Input 8'	'Prio 1'
	'Prio 2'	'Internal sine generator'	'Internal white noise generator'
	'Internal pink noise generator'	'Digital/Optical'	
['Zone']	'1' – '8'		
ZoneMute	'On'	'Off'	
['Zone']	'1' – '8'		
ZoneTreble	-9 to 9 in steps of 1 dB		
['Zone']	'1' – '8'		
ZoneVolume	-70 to 0 in steps of 1 dB		
['Zone']	'1' – '8'		

Cable and Adapter Requirements:

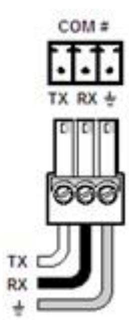
Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device:

Serial communication:

Port Type:	RS-232	**Parity:**	None
Baud Rate:	19200	**Stop Bits:**	One
Data Bits:	8	**Flow Control:**	None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND

General Notes:

Network communication:
When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	5001
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Yes
Port Changeability:	No

Ethernet Module Configuration Description:
Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Save None	#|R001|F001|SAVE|0|U|\x0D\x0A
Zone Bass 9 Zone 1	#|R001|F001|SB1|9.0|U|\x0D\x0A
Zone Bass -9 Zone 1	#|R001|F001|SB1|-9.0|U|\x0D\x0A
Zone Bass 9 Zone 2	#|R001|F001|SB2|9.0|U|\x0D\x0A
Zone Bass -9 Zone 2	#|R001|F001|SB2|-9.0|U|\x0D\x0A
Zone Bass 9 Zone 3	#|R001|F001|SB3|9.0|U|\x0D\x0A
Zone Bass -9 Zone 3	#|R001|F001|SB3|-9.0|U|\x0D\x0A
Zone Bass 9 Zone 4	#|R001|F001|SB4|9.0|U|\x0D\x0A
Zone Bass -9 Zone 4	#|R001|F001|SB4|-9.0|U|\x0D\x0A
Zone Bass 9 Zone 5	#|R001|F001|SB5|9.0|U|\x0D\x0A
Zone Bass -9 Zone 5	#|R001|F001|SB5|-9.0|U|\x0D\x0A
Zone Bass 9 Zone 6	#|R001|F001|SB6|9.0|U|\x0D\x0A
Zone Bass -9 Zone 6	#|R001|F001|SB6|-9.0|U|\x0D\x0A
Zone Bass 9 Zone 7	#|R001|F001|SB7|9.0|U|\x0D\x0A
Zone Bass -9 Zone 7	#|R001|F001|SB7|-9.0|U|\x0D\x0A
Zone Bass 9 Zone 8	#|R001|F001|SB8|9.0|U|\x0D\x0A
Zone Bass -9 Zone 8	#|R001|F001|SB8|-9.0|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone 1	#|R001|F001|SR1|11|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone 2	#|R001|F001|SR2|11|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone 3	#|R001|F001|SR3|11|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone 4	#|R001|F001|SR4|11|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone 5	#|R001|F001|SR5|11|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone 6	#|R001|F001|SR6|11|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone 7	#|R001|F001|SR7|11|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone 8	#|R001|F001|SR8|11|U|\x0D\x0A
Zone Input Routing Digital/Optical Zone All	#|R001|F001|SRALL|11^11^11^11^11^11^11^11|U|\x0D\x0A
Zone Input Routing Direct Input 1 Zone 1	#|R001|F001|SR1|1|U|\x0D\x0A
Zone Input Routing Direct Input 1 Zone 2	#|R001|F001|SR2|1|U|\x0D\x0A
Zone Input Routing Direct Input 1 Zone 3	#|R001|F001|SR3|1|U|\x0D\x0A
Zone Input Routing Direct Input 1 Zone 4	#|R001|F001|SR4|1|U|\x0D\x0A
Zone Input Routing Direct Input 1 Zone 5	#|R001|F001|SR5|1|U|\x0D\x0A

Zone Input Routing Direct Input 1 Zone 6	#|R001|F001|SR6|1|U|\x0D\x0A
Zone Input Routing Direct Input 1 Zone 7	#|R001|F001|SR7|1|U|\x0D\x0A
Zone Input Routing Direct Input 1 Zone 8	#|R001|F001|SR8|1|U|\x0D\x0A
Zone Input Routing Direct Input 1 Zone All	#|R001|F001|SRALL|1^1^1^1^1^1^1^1|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone 1	#|R001|F001|SR1|2|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone 2	#|R001|F001|SR2|2|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone 3	#|R001|F001|SR3|2|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone 4	#|R001|F001|SR4|2|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone 5	#|R001|F001|SR5|2|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone 6	#|R001|F001|SR6|2|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone 7	#|R001|F001|SR7|2|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone 8	#|R001|F001|SR8|2|U|\x0D\x0A
Zone Input Routing Direct Input 2 Zone All	#|R001|F001|SRALL|2^2^2^2^2^2^2^2|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone 1	#|R001|F001|SR1|3|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone 2	#|R001|F001|SR2|3|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone 3	#|R001|F001|SR3|3|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone 4	#|R001|F001|SR4|3|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone 5	#|R001|F001|SR5|3|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone 6	#|R001|F001|SR6|3|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone 7	#|R001|F001|SR7|3|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone 8	#|R001|F001|SR8|3|U|\x0D\x0A
Zone Input Routing Direct Input 3 Zone All	#|R001|F001|SRALL|3^3^3^3^3^3^3^3|U|\x0D\x0A
Zone Input Routing Direct Input 4 Zone 1	#|R001|F001|SR1|4|U|\x0D\x0A
Zone Input Routing Direct Input 4 Zone 2	#|R001|F001|SR2|4|U|\x0D\x0A

Zone Input Routing Direct Input 4 Zone 3	#|R001|F001|SR3|4|U|\x0D\x0A
Zone Input Routing Direct Input 4 Zone 4	#|R001|F001|SR4|4|U|\x0D\x0A
Zone Input Routing Direct Input 4 Zone 5	#|R001|F001|SR5|4|U|\x0D\x0A
Zone Input Routing Direct Input 4 Zone 6	#|R001|F001|SR6|4|U|\x0D\x0A
Zone Input Routing Direct Input 4 Zone 7	#|R001|F001|SR7|4|U|\x0D\x0A
Zone Input Routing Direct Input 4 Zone 8	#|R001|F001|SR8|4|U|\x0D\x0A
Zone Input Routing Direct Input 4 Zone All	#|R001|F001|SRALL|4^4^4^4^4^4^4^4|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone 1	#|R001|F001|SR1|5|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone 2	#|R001|F001|SR2|5|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone 3	#|R001|F001|SR3|5|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone 4	#|R001|F001|SR4|5|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone 5	#|R001|F001|SR5|5|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone 6	#|R001|F001|SR6|5|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone 7	#|R001|F001|SR7|5|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone 8	#|R001|F001|SR8|5|U|\x0D\x0A
Zone Input Routing Direct Input 5 Zone All	#|R001|F001|SRALL|5^5^5^5^5^5^5^5|U|\x0D\x0A
Zone Input Routing Direct Input 6 Zone 1	#|R001|F001|SR1|6|U|\x0D\x0A
Zone Input Routing Direct Input 6 Zone 2	#|R001|F001|SR2|6|U|\x0D\x0A
Zone Input Routing Direct Input 6 Zone 3	#|R001|F001|SR3|6|U|\x0D\x0A
Zone Input Routing Direct Input 6 Zone 4	#|R001|F001|SR4|6|U|\x0D\x0A
Zone Input Routing Direct Input 6 Zone 5	#|R001|F001|SR5|6|U|\x0D\x0A
Zone Input Routing Direct Input 6 Zone 6	#|R001|F001|SR6|6|U|\x0D\x0A
Zone Input Routing Direct Input 6 Zone 7	#|R001|F001|SR7|6|U|\x0D\x0A
Zone Input Routing Direct Input 6 Zone 8	#|R001|F001|SR8|6|U|\x0D\x0A

Zone Input Routing Direct Input 6 Zone All	#|R001|F001|SRALL|6^6^6^6^6^6^6^6|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone 1	#|R001|F001|SR1|7|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone 2	#|R001|F001|SR2|7|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone 3	#|R001|F001|SR3|7|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone 4	#|R001|F001|SR4|7|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone 5	#|R001|F001|SR5|7|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone 6	#|R001|F001|SR6|7|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone 7	#|R001|F001|SR7|7|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone 8	#|R001|F001|SR8|7|U|\x0D\x0A
Zone Input Routing Direct Input 7 Zone All	#|R001|F001|SRALL|7^7^7^7^7^7^7^7|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone 1	#|R001|F001|SR1|8|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone 2	#|R001|F001|SR2|8|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone 3	#|R001|F001|SR3|8|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone 4	#|R001|F001|SR4|8|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone 5	#|R001|F001|SR5|8|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone 6	#|R001|F001|SR6|8|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone 7	#|R001|F001|SR7|8|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone 8	#|R001|F001|SR8|8|U|\x0D\x0A
Zone Input Routing Direct Input 8 Zone All	#|R001|F001|SRALL|8^8^8^8^8^8^8^8|U|\x0D\x0A
Zone Input Routing Fiber Input 1 Zone 1	#|R001|F001|SR1|25|U|\x0D\x0A
Zone Input Routing Fiber Input 1 Zone 2	#|R001|F001|SR2|25|U|\x0D\x0A
Zone Input Routing Fiber Input 1 Zone 3	#|R001|F001|SR3|25|U|\x0D\x0A
Zone Input Routing Fiber Input 1 Zone 4	#|R001|F001|SR4|25|U|\x0D\x0A
Zone Input Routing Fiber Input 1 Zone 5	#|R001|F001|SR5|25|U|\x0D\x0A

Zone Input Routing Fiber Input 1 Zone 6	#|R001|F001|SR6|25|U|\x0D\x0A
Zone Input Routing Fiber Input 1 Zone 7	#|R001|F001|SR7|25|U|\x0D\x0A
Zone Input Routing Fiber Input 1 Zone 8	#|R001|F001|SR8|25|U|\x0D\x0A
Zone Input Routing Fiber Input 1 Zone All	#|R001|F001|SRALL|25^25^25^25^25^25^25^25|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone 1	#|R001|F001|SR1|26|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone 2	#|R001|F001|SR2|26|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone 3	#|R001|F001|SR3|26|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone 4	#|R001|F001|SR4|26|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone 5	#|R001|F001|SR5|26|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone 6	#|R001|F001|SR6|26|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone 7	#|R001|F001|SR7|26|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone 8	#|R001|F001|SR8|26|U|\x0D\x0A
Zone Input Routing Fiber Input 2 Zone All	#|R001|F001|SRALL|26^26^26^26^26^26^26^26|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone 1	#|R001|F001|SR1|27|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone 2	#|R001|F001|SR2|27|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone 3	#|R001|F001|SR3|27|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone 4	#|R001|F001|SR4|27|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone 5	#|R001|F001|SR5|27|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone 6	#|R001|F001|SR6|27|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone 7	#|R001|F001|SR7|27|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone 8	#|R001|F001|SR8|27|U|\x0D\x0A
Zone Input Routing Fiber Input 3 Zone All	#|R001|F001|SRALL|27^27^27^27^27^27^27^27|U|\x0D\x0A
Zone Input Routing Fiber Input 4 Zone 1	#|R001|F001|SR1|28|U|\x0D\x0A
Zone Input Routing Fiber Input 4 Zone 2	#|R001|F001|SR2|28|U|\x0D\x0A

Zone Input Routing Fiber Input 4 Zone 3	#|R001|F001|SR3|28|U|\x0D\x0A
Zone Input Routing Fiber Input 4 Zone 4	#|R001|F001|SR4|28|U|\x0D\x0A
Zone Input Routing Fiber Input 4 Zone 5	#|R001|F001|SR5|28|U|\x0D\x0A
Zone Input Routing Fiber Input 4 Zone 6	#|R001|F001|SR6|28|U|\x0D\x0A
Zone Input Routing Fiber Input 4 Zone 7	#|R001|F001|SR7|28|U|\x0D\x0A
Zone Input Routing Fiber Input 4 Zone 8	#|R001|F001|SR8|28|U|\x0D\x0A
Zone Input Routing Fiber Input 4 Zone All	#|R001|F001|SRALL|28^28^28^28^28^28^28^28|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone 1	#|R001|F001|SR1|29|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone 2	#|R001|F001|SR2|29|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone 3	#|R001|F001|SR3|29|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone 4	#|R001|F001|SR4|29|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone 5	#|R001|F001|SR5|29|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone 6	#|R001|F001|SR6|29|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone 7	#|R001|F001|SR7|29|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone 8	#|R001|F001|SR8|29|U|\x0D\x0A
Zone Input Routing Fiber Input 5 Zone All	#|R001|F001|SRALL|29^29^29^29^29^29^29^29|U|\x0D\x0A
Zone Input Routing Fiber Input 6 Zone 1	#|R001|F001|SR1|30|U|\x0D\x0A
Zone Input Routing Fiber Input 6 Zone 2	#|R001|F001|SR2|30|U|\x0D\x0A
Zone Input Routing Fiber Input 6 Zone 3	#|R001|F001|SR3|30|U|\x0D\x0A
Zone Input Routing Fiber Input 6 Zone 4	#|R001|F001|SR4|30|U|\x0D\x0A
Zone Input Routing Fiber Input 6 Zone 5	#|R001|F001|SR5|30|U|\x0D\x0A
Zone Input Routing Fiber Input 6 Zone 6	#|R001|F001|SR6|30|U|\x0D\x0A
Zone Input Routing Fiber Input 6 Zone 7	#|R001|F001|SR7|30|U|\x0D\x0A
Zone Input Routing Fiber Input 6 Zone 8	#|R001|F001|SR8|30|U|\x0D\x0A

Zone Input Routing Fiber Input 6 Zone All	#|R001|F001|SRALL|30^30^30^30^30^30^30^30|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone 1	#|R001|F001|SR1|31|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone 2	#|R001|F001|SR2|31|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone 3	#|R001|F001|SR3|31|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone 4	#|R001|F001|SR4|31|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone 5	#|R001|F001|SR5|31|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone 6	#|R001|F001|SR6|31|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone 7	#|R001|F001|SR7|31|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone 8	#|R001|F001|SR8|31|U|\x0D\x0A
Zone Input Routing Fiber Input 7 Zone All	#|R001|F001|SRALL|31^31^31^31^31^31^31^31|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone 1	#|R001|F001|SR1|32|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone 2	#|R001|F001|SR2|32|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone 3	#|R001|F001|SR3|32|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone 4	#|R001|F001|SR4|32|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone 5	#|R001|F001|SR5|32|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone 6	#|R001|F001|SR6|32|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone 7	#|R001|F001|SR7|32|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone 8	#|R001|F001|SR8|32|U|\x0D\x0A
Zone Input Routing Fiber Input 8 Zone All	#|R001|F001|SRALL|32^32^32^32^32^32^32^32|U|\x0D\x0A
Zone Input Routing Internal pink noise generator Zone 1	#|R001|F001|SR1|16|U|\x0D\x0A
Zone Input Routing Internal pink noise generator Zone 2	#|R001|F001|SR2|16|U|\x0D\x0A
Zone Input Routing Internal pink noise generator Zone 3	#|R001|F001|SR3|16|U|\x0D\x0A
Zone Input Routing Internal pink noise generator Zone 4	#|R001|F001|SR4|16|U|\x0D\x0A
Zone Input Routing Internal pink noise generator Zone 5	#|R001|F001|SR5|16|U|\x0D\x0A

Zone Input Routing Internal pink noise generator Zone 6	#|R001|F001|SR6|16|U|\x0D\x0A
Zone Input Routing Internal pink noise generator Zone 7	#|R001|F001|SR7|16|U|\x0D\x0A
Zone Input Routing Internal pink noise generator Zone 8	#|R001|F001|SR8|16|U|\x0D\x0A
Zone Input Routing Internal pink noise generator Zone All	#|R001|F001|SRALL|16^16^16^16^16^16^16^16|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone 1	#|R001|F001|SR1|14|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone 2	#|R001|F001|SR2|14|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone 3	#|R001|F001|SR3|14|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone 4	#|R001|F001|SR4|14|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone 5	#|R001|F001|SR5|14|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone 6	#|R001|F001|SR6|14|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone 7	#|R001|F001|SR7|14|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone 8	#|R001|F001|SR8|14|U|\x0D\x0A
Zone Input Routing Internal sine generator Zone All	#|R001|F001|SRALL|14^14^14^14^14^14^14^14|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone 1	#|R001|F001|SR1|15|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone 2	#|R001|F001|SR2|15|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone 3	#|R001|F001|SR3|15|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone 4	#|R001|F001|SR4|15|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone 5	#|R001|F001|SR5|15|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone 6	#|R001|F001|SR6|15|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone 7	#|R001|F001|SR7|15|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone 8	#|R001|F001|SR8|15|U|\x0D\x0A
Zone Input Routing Internal white noise generator Zone All	#|R001|F001|SRALL|15^15^15^15^15^15^15^15|U|\x0D\x0A
Zone Input Routing Prio 1 Zone 1	#|R001|F001|SR1|12|U|\x0D\x0A
Zone Input Routing Prio 1 Zone 2	#|R001|F001|SR2|12|U|\x0D\x0A

Zone Input Routing Prio 1 Zone 3	#|R001|F001|SR3|12|U|\x0D\x0A
Zone Input Routing Prio 1 Zone 4	#|R001|F001|SR4|12|U|\x0D\x0A
Zone Input Routing Prio 1 Zone 5	#|R001|F001|SR5|12|U|\x0D\x0A
Zone Input Routing Prio 1 Zone 6	#|R001|F001|SR6|12|U|\x0D\x0A
Zone Input Routing Prio 1 Zone 7	#|R001|F001|SR7|12|U|\x0D\x0A
Zone Input Routing Prio 1 Zone 8	#|R001|F001|SR8|12|U|\x0D\x0A
Zone Input Routing Prio 1 Zone All	#|R001|F001|SRALL|12^12^12^12^12^12^12^12|U|\x0D\x0A
Zone Input Routing Prio 2 Zone 1	#|R001|F001|SR1|13|U|\x0D\x0A
Zone Input Routing Prio 2 Zone 2	#|R001|F001|SR2|13|U|\x0D\x0A
Zone Input Routing Prio 2 Zone 3	#|R001|F001|SR3|13|U|\x0D\x0A
Zone Input Routing Prio 2 Zone 4	#|R001|F001|SR4|13|U|\x0D\x0A
Zone Input Routing Prio 2 Zone 5	#|R001|F001|SR5|13|U|\x0D\x0A
Zone Input Routing Prio 2 Zone 6	#|R001|F001|SR6|13|U|\x0D\x0A
Zone Input Routing Prio 2 Zone 7	#|R001|F001|SR7|13|U|\x0D\x0A
Zone Input Routing Prio 2 Zone 8	#|R001|F001|SR8|13|U|\x0D\x0A
Zone Input Routing Prio 2 Zone All	#|R001|F001|SRALL|13^13^13^13^13^13^13^13|U|\x0D\x0A
Zone Input Routing Wall Plate Input 1 Zone 1	#|R001|F001|SR1|17|U|\x0D\x0A
Zone Input Routing Wall Plate Input 1 Zone 2	#|R001|F001|SR2|17|U|\x0D\x0A
Zone Input Routing Wall Plate Input 1 Zone 3	#|R001|F001|SR3|17|U|\x0D\x0A
Zone Input Routing Wall Plate Input 1 Zone 4	#|R001|F001|SR4|17|U|\x0D\x0A
Zone Input Routing Wall Plate Input 1 Zone 5	#|R001|F001|SR5|17|U|\x0D\x0A
Zone Input Routing Wall Plate Input 1 Zone 6	#|R001|F001|SR6|17|U|\x0D\x0A
Zone Input Routing Wall Plate Input 1 Zone 7	#|R001|F001|SR7|17|U|\x0D\x0A
Zone Input Routing Wall Plate Input 1 Zone 8	#|R001|F001|SR8|17|U|\x0D\x0A

Zone Input Routing Wall Plate Input 1 Zone All	#|R001|F001|SRALL|17^17^17^17^17^17^17^17|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone 1	#|R001|F001|SR1|18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone 2	#|R001|F001|SR2|18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone 3	#|R001|F001|SR3|18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone 4	#|R001|F001|SR4|18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone 5	#|R001|F001|SR5|18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone 6	#|R001|F001|SR6|18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone 7	#|R001|F001|SR7|18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone 8	#|R001|F001|SR8|18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 2 Zone All	#|R001|F001|SRALL|18^18^18^18^18^18^18^18|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone 1	#|R001|F001|SR1|19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone 2	#|R001|F001|SR2|19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone 3	#|R001|F001|SR3|19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone 4	#|R001|F001|SR4|19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone 5	#|R001|F001|SR5|19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone 6	#|R001|F001|SR6|19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone 7	#|R001|F001|SR7|19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone 8	#|R001|F001|SR8|19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 3 Zone All	#|R001|F001|SRALL|19^19^19^19^19^19^19^19|U|\x0D\x0A
Zone Input Routing Wall Plate Input 4 Zone 1	#|R001|F001|SR1|20|U|\x0D\x0A
Zone Input Routing Wall Plate Input 4 Zone 2	#|R001|F001|SR2|20|U|\x0D\x0A
Zone Input Routing Wall Plate Input 4 Zone 3	#|R001|F001|SR3|20|U|\x0D\x0A
Zone Input Routing Wall Plate Input 4 Zone 4	#|R001|F001|SR4|20|U|\x0D\x0A
Zone Input Routing Wall Plate Input 4 Zone 5	#|R001|F001|SR5|20|U|\x0D\x0A

Zone Input Routing Wall Plate Input 4 Zone 6	#|R001|F001|SR6|20|U|\x0D\x0A
Zone Input Routing Wall Plate Input 4 Zone 7	#|R001|F001|SR7|20|U|\x0D\x0A
Zone Input Routing Wall Plate Input 4 Zone 8	#|R001|F001|SR8|20|U|\x0D\x0A
Zone Input Routing Wall Plate Input 4 Zone All	#|R001|F001|SRALL|20^20^20^20^20^20^20^20|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone 1	#|R001|F001|SR1|21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone 2	#|R001|F001|SR2|21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone 3	#|R001|F001|SR3|21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone 4	#|R001|F001|SR4|21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone 5	#|R001|F001|SR5|21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone 6	#|R001|F001|SR6|21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone 7	#|R001|F001|SR7|21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone 8	#|R001|F001|SR8|21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 5 Zone All	#|R001|F001|SRALL|21^21^21^21^21^21^21^21|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone 1	#|R001|F001|SR1|22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone 2	#|R001|F001|SR2|22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone 3	#|R001|F001|SR3|22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone 4	#|R001|F001|SR4|22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone 5	#|R001|F001|SR5|22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone 6	#|R001|F001|SR6|22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone 7	#|R001|F001|SR7|22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone 8	#|R001|F001|SR8|22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 6 Zone All	#|R001|F001|SRALL|22^22^22^22^22^22^22^22|U|\x0D\x0A
Zone Input Routing Wall Plate Input 7 Zone 1	#|R001|F001|SR1|23|U|\x0D\x0A
Zone Input Routing Wall Plate Input 7 Zone 2	#|R001|F001|SR2|23|U|\x0D\x0A

Zone Input Routing Wall Plate Input 7 Zone 3	#|R001|F001|SR3|23|U|\x0D\x0A
Zone Input Routing Wall Plate Input 7 Zone 4	#|R001|F001|SR4|23|U|\x0D\x0A
Zone Input Routing Wall Plate Input 7 Zone 5	#|R001|F001|SR5|23|U|\x0D\x0A
Zone Input Routing Wall Plate Input 7 Zone 6	#|R001|F001|SR6|23|U|\x0D\x0A
Zone Input Routing Wall Plate Input 7 Zone 7	#|R001|F001|SR7|23|U|\x0D\x0A
Zone Input Routing Wall Plate Input 7 Zone 8	#|R001|F001|SR8|23|U|\x0D\x0A
Zone Input Routing Wall Plate Input 7 Zone All	#|R001|F001|SRALL|23^23^23^23^23^23^23^23|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone 1	#|R001|F001|SR1|24|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone 2	#|R001|F001|SR2|24|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone 3	#|R001|F001|SR3|24|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone 4	#|R001|F001|SR4|24|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone 5	#|R001|F001|SR5|24|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone 6	#|R001|F001|SR6|24|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone 7	#|R001|F001|SR7|24|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone 8	#|R001|F001|SR8|24|U|\x0D\x0A
Zone Input Routing Wall Plate Input 8 Zone All	#|R001|F001|SRALL|24^24^24^24^24^24^24^24|U|\x0D\x0A
Zone Mute Off Zone 1	#|R001|F001|SM1|0|U|\x0D\x0A
Zone Mute Off Zone 2	#|R001|F001|SM2|0|U|\x0D\x0A
Zone Mute Off Zone 3	#|R001|F001|SM3|0|U|\x0D\x0A
Zone Mute Off Zone 4	#|R001|F001|SM4|0|U|\x0D\x0A
Zone Mute Off Zone 5	#|R001|F001|SM5|0|U|\x0D\x0A
Zone Mute Off Zone 6	#|R001|F001|SM6|0|U|\x0D\x0A
Zone Mute Off Zone 7	#|R001|F001|SM7|0|U|\x0D\x0A
Zone Mute Off Zone 8	#|R001|F001|SM8|0|U|\x0D\x0A
Zone Mute Off Zone All	#|R001|F001|SMALL|0^0^0^0^0^0^0^0|U|\x0D\x0A
Zone Mute On Zone 1	#|R001|F001|SM1|1|U|\x0D\x0A
Zone Mute On Zone 2	#|R001|F001|SM2|1|U|\x0D\x0A
Zone Mute On Zone 3	#|R001|F001|SM3|1|U|\x0D\x0A
Zone Mute On Zone 4	#|R001|F001|SM4|1|U|\x0D\x0A
Zone Mute On Zone 5	#|R001|F001|SM5|1|U|\x0D\x0A
Zone Mute On Zone 6	#|R001|F001|SM6|1|U|\x0D\x0A
Zone Mute On Zone 7	#|R001|F001|SM7|1|U|\x0D\x0A
Zone Mute On Zone 8	#|R001|F001|SM8|1|U|\x0D\x0A

Zone Mute On Zone All	#|R001|F001|SMALL|1^1^1^1^1^1^1^1|U|\x0D\x0A
Zone Treble 9 Zone 1	#|R001|F001|ST1|9|U|\x0D\x0A
Zone Treble -9 Zone 1	#|R001|F001|ST1|-9|U|\x0D\x0A
Zone Treble 9 Zone 2	#|R001|F001|ST2|9|U|\x0D\x0A
Zone Treble -9 Zone 2	#|R001|F001|ST2|-9|U|\x0D\x0A
Zone Treble 9 Zone 3	#|R001|F001|ST3|9|U|\x0D\x0A
Zone Treble -9 Zone 3	#|R001|F001|ST3|-9|U|\x0D\x0A
Zone Treble 9 Zone 4	#|R001|F001|ST4|9|U|\x0D\x0A
Zone Treble -9 Zone 4	#|R001|F001|ST4|-9|U|\x0D\x0A
Zone Treble 9 Zone 5	#|R001|F001|ST5|9|U|\x0D\x0A
Zone Treble -9 Zone 5	#|R001|F001|ST5|-9|U|\x0D\x0A
Zone Treble 9 Zone 6	#|R001|F001|ST6|9|U|\x0D\x0A
Zone Treble -9 Zone 6	#|R001|F001|ST6|-9|U|\x0D\x0A
Zone Treble 9 Zone 7	#|R001|F001|ST7|9|U|\x0D\x0A
Zone Treble -9 Zone 7	#|R001|F001|ST7|-9|U|\x0D\x0A
Zone Treble 9 Zone 8	#|R001|F001|ST8|9|U|\x0D\x0A
Zone Treble -9 Zone 8	#|R001|F001|ST8|-9|U|\x0D\x0A
Zone Volume 0 Zone 1	#|R001|F001|SV1|-0.0|U|\x0D\x0A
Zone Volume 0 Zone 2	#|R001|F001|SV2|-0.0|U|\x0D\x0A
Zone Volume 0 Zone 3	#|R001|F001|SV3|-0.0|U|\x0D\x0A
Zone Volume 0 Zone 4	#|R001|F001|SV4|-0.0|U|\x0D\x0A
Zone Volume 0 Zone 5	#|R001|F001|SV5|-0.0|U|\x0D\x0A
Zone Volume 0 Zone 6	#|R001|F001|SV6|-0.0|U|\x0D\x0A
Zone Volume 0 Zone 7	#|R001|F001|SV7|-0.0|U|\x0D\x0A
Zone Volume 0 Zone 8	#|R001|F001|SV8|-0.0|U|\x0D\x0A
Zone Volume 0 Zone All	#|R001|F001|SVALL|-0.0^-0.0^-0.0^-0.0^-0.0^-0.0^-0.0^-0.0|U|\x0D\x0A
Zone Volume -70 Zone 1	#|R001|F001|SV1|70.0|U|\x0D\x0A
Zone Volume -70 Zone 2	#|R001|F001|SV2|70.0|U|\x0D\x0A
Zone Volume -70 Zone 3	#|R001|F001|SV3|70.0|U|\x0D\x0A
Zone Volume -70 Zone 4	#|R001|F001|SV4|70.0|U|\x0D\x0A
Zone Volume -70 Zone 5	#|R001|F001|SV5|70.0|U|\x0D\x0A
Zone Volume -70 Zone 6	#|R001|F001|SV6|70.0|U|\x0D\x0A
Zone Volume -70 Zone 7	#|R001|F001|SV7|70.0|U|\x0D\x0A
Zone Volume -70 Zone 8	#|R001|F001|SV8|70.0|U|\x0D\x0A
Zone Volume -70 Zone All	#|R001|F001|SVALL|70.0^70.0^70.0^70.0^70.0^70.0^70.0^70.0|U|\x0D\x0A

Appendix B. Query Commands

Zone Bass Zone 1	#|R001|F001|GB1|0|U|\x0D\x0A
Zone Bass Zone 2	#|R001|F001|GB2|0|U|\x0D\x0A
Zone Bass Zone 3	#|R001|F001|GB3|0|U|\x0D\x0A
Zone Bass Zone 4	#|R001|F001|GB4|0|U|\x0D\x0A
Zone Bass Zone 5	#|R001|F001|GB5|0|U|\x0D\x0A
Zone Bass Zone 6	#|R001|F001|GB6|0|U|\x0D\x0A
Zone Bass Zone 7	#|R001|F001|GB7|0|U|\x0D\x0A
Zone Bass Zone 8	#|R001|F001|GB8|0|U|\x0D\x0A

Zone Input Routing Zone 1	#|R001|F001|GR1|0|U|\x0D\x0A
Zone Input Routing Zone 2	#|R001|F001|GR2|0|U|\x0D\x0A
Zone Input Routing Zone 3	#|R001|F001|GR3|0|U|\x0D\x0A
Zone Input Routing Zone 4	#|R001|F001|GR4|0|U|\x0D\x0A
Zone Input Routing Zone 5	#|R001|F001|GR5|0|U|\x0D\x0A
Zone Input Routing Zone 6	#|R001|F001|GR6|0|U|\x0D\x0A
Zone Input Routing Zone 7	#|R001|F001|GR7|0|U|\x0D\x0A
Zone Input Routing Zone 8	#|R001|F001|GR8|0|U|\x0D\x0A
Zone Mute Zone 1	#|R001|F001|GM1|0|U|\x0D\x0A
Zone Mute Zone 2	#|R001|F001|GM2|0|U|\x0D\x0A
Zone Mute Zone 3	#|R001|F001|GM3|0|U|\x0D\x0A
Zone Mute Zone 4	#|R001|F001|GM4|0|U|\x0D\x0A
Zone Mute Zone 5	#|R001|F001|GM5|0|U|\x0D\x0A
Zone Mute Zone 6	#|R001|F001|GM6|0|U|\x0D\x0A
Zone Mute Zone 7	#|R001|F001|GM7|0|U|\x0D\x0A
Zone Mute Zone 8	#|R001|F001|GM8|0|U|\x0D\x0A
Zone Treble Zone 1	#|R001|F001|GT1|0|U|\x0D\x0A
Zone Treble Zone 2	#|R001|F001|GT2|0|U|\x0D\x0A
Zone Treble Zone 3	#|R001|F001|GT3|0|U|\x0D\x0A
Zone Treble Zone 4	#|R001|F001|GT4|0|U|\x0D\x0A
Zone Treble Zone 5	#|R001|F001|GT5|0|U|\x0D\x0A
Zone Treble Zone 6	#|R001|F001|GT6|0|U|\x0D\x0A
Zone Treble Zone 7	#|R001|F001|GT7|0|U|\x0D\x0A
Zone Treble Zone 8	#|R001|F001|GT8|0|U|\x0D\x0A
Zone Volume Zone 1	#|R001|F001|GV1|0|U|\x0D\x0A
Zone Volume Zone 2	#|R001|F001|GV2|0|U|\x0D\x0A
Zone Volume Zone 3	#|R001|F001|GV3|0|U|\x0D\x0A
Zone Volume Zone 4	#|R001|F001|GV4|0|U|\x0D\x0A
Zone Volume Zone 5	#|R001|F001|GV5|0|U|\x0D\x0A
Zone Volume Zone 6	#|R001|F001|GV6|0|U|\x0D\x0A
Zone Volume Zone 7	#|R001|F001|GV7|0|U|\x0D\x0A
Zone Volume Zone 8	#|R001|F001|GV8|0|U|\x0D\x0A